

Does Syntactic Structure Make Language Models More Robust to Noise?

LSTM vs. RNN Subject-Verb Agreement Under Training-Data Corruption

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Abstract

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We evaluate every model on the same leak-free set of 19,819 minimal pairs (1,528 with an attractor, 18,291 without), held identical across all five noise conditions. Unless noted, reported values are the mean over six LSTM runs per condition (seeds {1, 2, 3}, run on both local MPS and Colab T4); $\pm$ denotes standard deviation across runs. Results for the RNN arm are forthcoming and will be added to Table 1 once training completes.

5.1 Noise degrades agreement, with a threshold near 50%

Overall minimal-pair accuracy is remarkably stable across the first four noise levels and then collapses at the highest rate (Table 1, Figure 1). Accuracy falls only 2.7 points as the error rate climbs from 0% to 10% ($0.961 \rightarrow 0.934$), but drops 22.6 points between 10% and 50% ($0.934 \rightarrow 0.708$). The model thus tolerates substantial agreement noise—up to roughly one corrupted verb in ten—before its grasp of the rule breaks down. This non-linear, threshold-like profile matches the first outcome anticipated in our proposal: input quality can be degraded considerably before grammar learning is

Condition	Rate	Acc. (all)	Attractor gap
baseline	0.000	0.961 ± 0.006	0.073 ± 0.023
low	0.004	0.958 ± 0.009	0.087 ± 0.030
medium-low	0.020	0.952 ± 0.014	0.103 ± 0.038
medium-high	0.100	0.934 ± 0.020	0.153 ± 0.051
high	0.500	0.708 ± 0.018	0.198 ± 0.033

Table 1: LSTM minimal-pair accuracy and attractor gap by training-noise rate (mean $\pm$ std over six runs). Accuracy is stable through 10% noise and collapses at 50%; the attractor gap grows monotonically throughout.

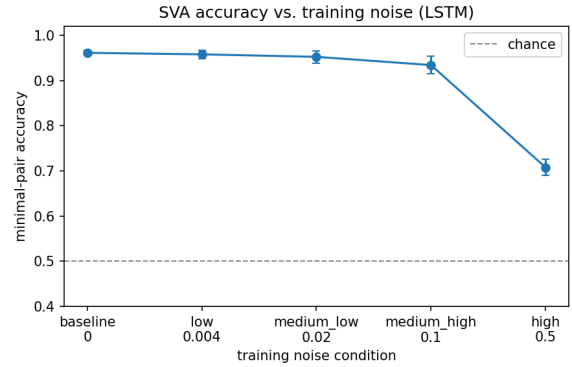


Figure 1: Overall minimal-pair accuracy vs. injected SVA noise rate. The dashed line marks chance (0.5). Performance is flat through 10% noise and falls sharply at 50%.

affected, but beyond a critical point performance falls sharply toward chance.

5.2 Noise pushes the model toward the linear rule

Splitting accuracy by attractor presence isolates *which* rule the model relies on (Figure 2). No-attractor items—where the hierarchical and linear rules make the same prediction—stay above 0.94 until the 50% condition. Attractor items, where the two rules disagree, erode far faster: accuracy falls from 0.894 at baseline to 0.525 (near chance) at 50% noise. As a result the *attractor gap*—our index of reliance on the linear, nearest-noun

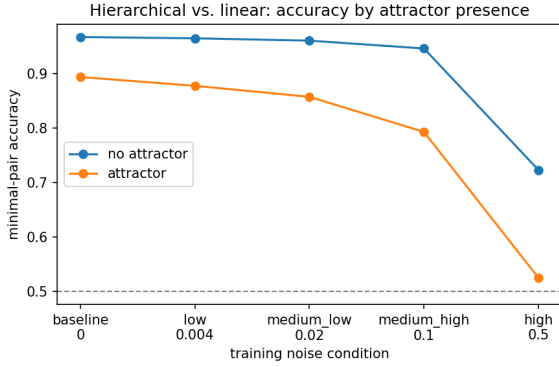


Figure 2: Accuracy on attractor vs. no-attractor items. The two cells diverge as noise rises: attractor accuracy approaches chance while no-attractor accuracy stays high until 50%.

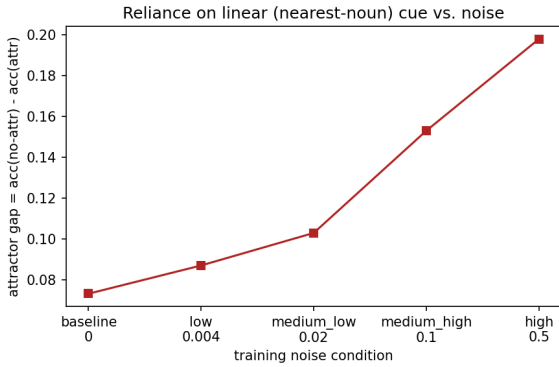


Figure 3: Attractor gap (no-attractor minus attractor accuracy) vs. noise rate. The monotonic rise indicates increasing reliance on the linear cue.

heuristic—grows monotonically with noise, from 0.073 to 0.198 (Figure 3). Noisier training data drives the model away from the hierarchical rule and toward the linear shortcut, consistent with the “noise pushes models toward simpler generalization” branch of our second predicted outcome.

5.3 Learning trajectories: delay vs. prevention

To ask *when*—not just whether—the rule is acquired, we checkpoint every 500 optimizer steps and re-score the full minimal-pair set at each point (Figure 4). The curves separate the two ways noise can hurt. For the baseline through medium-high conditions, accuracy follows nearly the same trajectory and converges to roughly the same endpoint (0.95–0.97): noise *delays* acquisition but does not prevent it. The 50% condition behaves qualitatively differently—it plateaus near 0.69 and never reaches the rule, i.e. *prevention*. The attractor gap

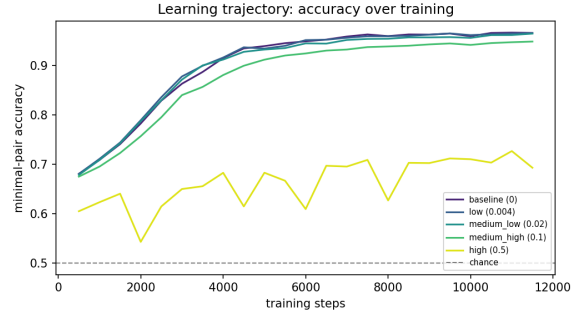


Figure 4: Minimal-pair accuracy over training steps, one curve per noise condition. Low-to-medium noise tracks the clean trajectory and converges (delay); the 50% condition plateaus well below the others (prevention).

reinforces this: for the clean model the gap shrinks over training as the hierarchical rule consolidates (0.148 $\rightarrow$ 0.054), whereas for the 50% model it *grows* with more training (0.143 $\rightarrow$ 0.197)—additional exposure to corrupted data entrenches the linear heuristic rather than correcting it.

6 Discussion

When does the rule break? Two findings constrain when subject–verb agreement survives noisy training. First, the breakdown is *threshold-like*, not gradual: accuracy is nearly flat up to a 10% corruption rate and only collapses at 50% (Table 1). Small amounts of ungrammatical input are absorbed, much as a child’s grammar is not derailed by occasional speech errors. Second, the attractor gap is a more sensitive early-warning signal than overall accuracy: it climbs steadily (0.073 $\rightarrow$ 0.198) even while aggregate accuracy looks healthy, revealing that the model is quietly trading the hierarchical rule for the linear nearest-noun heuristic well before its overall performance gives way. The trajectory analysis sharpens the distinction between the two ways noise can act. Up to 10% noise, learning is merely *delayed*—every condition converges to the same accuracy given enough steps—whereas at 50% it is *prevented*: the model plateaus far short of the rule, and further training deepens rather than repairs its reliance on the linear cue.

Does structure buy robustness? Our central architectural question—whether a syntactically biased model (RNNG) withstands noise better than a purely sequential one (LSTM)—requires the RNNG runs to answer, and those are still in progress (Section 5). The LSTM results establish

the baseline against which that comparison will be read: a recurrent model with no explicit hierarchical bias acquires the hierarchical rule cleanly under low noise but is pushed toward the linear heuristic as noise grows, and fails outright at 50%. If the RNNG preserves the hierarchical rule at noise levels where the LSTM has already defaulted to the linear shortcut, that would implicate an architectural structural prior as a source of robustness. If both architectures degrade together, it would instead suggest the limiting factor is the statistical quality of the input rather than the model’s inductive bias—bearing directly on the poverty-of-the-stimulus framing of Section 1.

Cognitive-modeling angle. The pattern is suggestive for how structural knowledge might arise from statistical learning. The LSTM does not begin agreement-aware; the hierarchical rule is *acquired* over training, and in the clean setting the attractor gap shrinks as that rule consolidates. Noise does not flip a switch so much as shift the balance between two competing generalizations—hierarchical and linear—both of which the data can support. Heavy enough noise makes the linear rule the better fit to the (corrupted) input, and the model follows the data. That a distributional learner recovers the hierarchical rule at all from this templatic corpus is itself evidence that the structure is more learnable from input than a strong nativist account would predict; that it abandons the rule under sufficient noise marks the limit of that learnability.

Threats to validity. Three caveats bound these claims; we treat them fully in Section 7. Our noise is synthetic and uniform—a fixed per-sentence flip probability—rather than the structured, frequency-dependent errors of natural input. The corpus is templatic and vocabulary-controlled, which aids control but limits external validity. Finally, the trajectory analysis is single-seed; the converging-versus-plateauing pattern is clear, but the precise crossover noise rate should be read as approximate until replicated across seeds.

7 Conclusion

Limitations

References

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A Reproducibility Details

B Additional Figures